Column 1	Work package	Owner	# Effort (person-hours)	Method / assumptions
1.1.1	Define the problem	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration (~2 person-hours/day); concise 1-2 page preliminary problem statement.
1.1.2	Identify target users and stakeholders	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; two primary end-user roles (Teacher, Student).
1.1.3	Define the scope and must-have features	Lê Đoàn Nhật Huy	4	Three-point estimate: O=2h, M=4h, P=6h -> E=4h (2-day duration); prioritizing 12 features (F01-F12) using MoSCoW.
1.2.1	Propose the preliminary application architecture	Trần Hoàng Minh Khang	2	Analogy: 1-day duration; similar to C4 System Context diagrams in previous coursework.
1.2.2	Propose the technology stack	Trần Hoàng Minh Khang	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); evaluating Nuxt 4, Drizzle vs Prisma, and Vitest/Playwright.
1.2.3	Propose the hosting / deployment environment	Trần Hoàng Minh Khang	2	Expert judgement: 1-day duration; comparing Cloudflare Pages/Workers, Vercel, and Render.
1.3.1	Define project phases and exit criteria	Huỳnh Trung Ngạn	4	Expert judgement: 2-day duration; defining exit criteria across 5 Waterfall phase gates (PA1-PA5).
1.3.2	Build a work breakdown structure (WBS)	Huỳnh Trung Ngạn	8	Three-point estimate: O=6h, M=8h, P=10h -> E=8h (4-day duration); decomposing 5 project phases into 3 levels.
1.3.3	Assign roles and responsibilities	Huỳnh Trung Ngạn	2	Expert judgement: 1-day duration; mapping 4 team members to primary and backup roles.
1.3.4	Present project schedule	Trương Hiến Long	4	Analogy: 2-day duration; building a 10-week Gantt chart and critical path similar to standard course templates.
1.3.5	Present work package effort estimation	Trương Hiến Long	4	Expert judgement: 2-day duration; calculating person-hours and documenting estimation methods for all WBS items.
1.4.1	Identify technology risks	Trần Hoàng Minh Khang	2	Expert judgement: 1-day duration; assessing Cloudflare V8 Isolate limits and PostgreSQL performance risks.
1.4.2	Identify requirements risks	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; assessing scope creep, estimation, and key member availability risks.
1.5.1	Set up communication channels	Huỳnh Trung Ngạn	2	Analogy: 1-day duration; standard Discord server setup and TA onboarding.
1.5.2	Set up task board	Huỳnh Trung Ngạn	4	Analogy: 2-day duration; configuring GitHub Projects board and creating issues for WBS work packages.
1.5.3	Set up collaboration tools	Trương Hiến Long	4	Analogy: 2-day duration; initializing GitHub repository, branch protection rules, and folder structure.
2.1.1	Define the functional requirements	Lê Đoàn Nhật Huy	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); specifying user requirements for F01-F09.
2.1.2	Define the non-functional requirements	Lê Đoàn Nhật Huy	4	Expert judgement: 2-day duration; specifying performance, security, and PWA/SPA compatibility targets.
2.1.3	Assign priorities to the requirements	Huỳnh Trung Ngạn	2	Expert judgement: 1-day duration; reviewing and refining MoSCoW priorities with stakeholders.
2.1.4	Define business rules	Lê Đoàn Nhật Huy	4	Expert judgement: 2-day duration; defining role permissions, enrollment constraints, and submission policies.
2.2.1	List use cases	Lê Đoàn Nhật Huy	8	Three-point estimate: O=6h, M=8h, P=10h -> E=8h (4-day duration); writing detailed use case specifications and flows.
2.2.2	List actors	Trần Hoàng Minh Khang	4	Analogy: 2-day duration; identifying actors and drawing the complete UML Use Case diagram.
2.3.1	Deliver a prototype / mockups	Trần Hoàng Minh Khang	8	Three-point estimate: O=6h, M=8h, P=10h -> E=8h (4-day duration); designing low/mid-fidelity wireframes for main workflows.
2.4.1	Make an RTM (Requirements Traceability Matrix)	Huỳnh Trung Ngạn	8	Analogy: 4-day duration; cross-referencing functional requirements, use cases, and design/test artifacts.
3.1.1	Decide on the architectural style	Trần Hoàng Minh Khang	2	Expert judgement: 1-day duration; documenting ADR for Nuxt 4 full-stack modular architecture on Edge.
3.1.2	Draw the system context diagram (update from PA1)	Trần Hoàng Minh Khang	4	Analogy: 2-day duration; refining PA1 C4 Level 1 diagram based on finalized PA2 SRS.
3.1.3	Draw the container diagram	Trần Hoàng Minh Khang	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); modeling C4 Level 2 (SPA, Nitro API, PostgreSQL, R2).
3.1.4	Draw the component diagram	Trần Hoàng Minh Khang	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); modeling C4 Level 3 backend and frontend modules.
3.1.5	Draw the deployment Diagram	Trương Hiến Long	4	Expert judgement: 2-day duration; mapping Cloudflare Pages/Workers, R2, and managed PostgreSQL nodes.
3.2.1	Draw the conceptual class diagram	Lê Đoàn Nhật Huy	4	Analogy: 2-day duration; domain modeling of core entities (User, Course, Enrollment, Material, Assignment).
3.2.2	Draw the design-level class diagram	Trần Hoàng Minh Khang	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); specifying methods, types, and relationships.
3.2.3	Draw the sequence diagrams for the main flow	Huỳnh Trung Ngạn	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); modeling interactions for key use cases.
3.2.4	Draw the State Machine Diagrams	Huỳnh Trung Ngạn	4	Analogy: 2-day duration; modeling lifecycle states for Courses and Assignment Submissions.
3.2.5	Define Design Principles and Patterns	Trương Hiến Long	4	Expert judgement: 2-day duration; specifying SOLID principles, repository/composable patterns, and middleware.
3.3.1	Draw the conceptual ERD of the stored data	Trương Hiến Long	4	Analogy: 2-day duration; entity-relationship modeling aligned with conceptual class diagram.
3.3.2	Schema Design & Modeling	Trương Hiến Long	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); defining Drizzle ORM PostgreSQL physical schema.
3.3.3	List Data specifications for DBMS	Trương Hiến Long	4	Expert judgement: 2-day duration; documenting column types, constraints, and foreign key policies.
3.3.4	Identify normal form and the indexes created for frequent queries	Trương Hiến Long	2	Expert judgement: 1-day duration; verifying 3NF/BCNF compliance and indexing foreign keys (mitigating Risk R5).
3.4.1	Draw the screen flow diagram (navigation & routing)	Trần Hoàng Minh Khang	4	Analogy: 2-day duration; mapping Nuxt page routing and role-based navigation guards.
3.4.2	Design page views and layout (wireframes)	Trần Hoàng Minh Khang	8	Three-point estimate: O=6h, M=8h, P=10h -> E=8h (4-day duration); high-fidelity UI layouts using Nuxt UI/Tailwind conventions.
3.4.3	List UX requirements (Accessibility)	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; defining WCAG contrast, keyboard navigation, and responsive design rules.
4.1.1	List features to be tested or not tested	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; scoping test coverage around Must-have features (F01-F05).
4.1.2	Map each feature with a test technique	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; assigning Equivalence Partitioning (EP), BVA, and Decision Tables.
4.1.3	Specify test environment and test data requirements	Trần Hoàng Minh Khang	6	Three-point estimate: O=4h, M=6h, P=8h -> E=6h (3-day duration); defining Vitest, Playwright, and local DB/R2 mock setups.

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4.2.1	List test cases for each feature/function	Lê Đoàn Nhật Huy	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); writing repeatable test cases with preconditions and expected results.
4.3.1	Map every Must-Have feature/requirement to >= 1 test case	Huỳnh Trung Ngạn	2	Analogy: 1-day duration; updating RTM to link F01-F05 requirements to PA4 test cases.
5.1.1	Set up and verify the coding environment and toolings	Trương Hiển Long	2	Analogy: 1-day duration; initializing Nuxt 4, TypeScript, Tailwind CSS, and Wrangler CLI.
5.1.2	Set up the testing environment	Trương Hiển Long	2	Analogy: 1-day duration; configuring @nuxt/test-utils, Vitest, and Playwright runners.
5.1.3	Document coding standards and conventions	Huỳnh Trung Ngạn	2	Expert judgement: 1-day duration; configuring ESLint, Prettier, and Conventional Commits guidelines.
5.1.4	Configure continuous integration pipeline	Trương Hiển Long	2	Analogy: 1-day duration; setting up GitHub Actions workflow for automated linting and testing.
5.2.1	Implement navigation and routing (according to screen flow diagram)	Trần Hoàng Minh Khang	4	Analogy: 2-day duration; building Nuxt pages, layouts, and auth middleware.
5.2.2	Implement page views and layout (according to wireframes)	Trần Hoàng Minh Khang	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); building responsive components with Nuxt UI.
5.2.3	Verify / Implement accessibility (according to UX requirements)	Trần Hoàng Minh Khang	2	Expert judgement: 1-day duration; auditing ARIA attributes, semantic HTML, and keyboard focus.
5.2.4	Wire CTAs to backend API endpoints	Huỳnh Trung Ngạn	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); connecting frontend forms/views to Nitro API via useFetch/$fetch.
5.3.1	Install database engine	Trương Hiển Long	4	Analogy: 2-day duration; provisioning managed PostgreSQL (Neon/Supabase) and local Docker PostgreSQL 16.
5.3.2	Configure access controls and credentials	Trương Hiển Long	2	Expert judgement: 1-day duration; setting up environment variables, connection pooling, and role permissions.
5.3.3	Implement database schema and tables (according to schema design)	Trương Hiển Long	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); writing Drizzle ORM schema definitions and running migrations.
5.3.4	Seed test data	Lê Đoàn Nhật Huy	2	Analogy: 1-day duration; writing seed scripts for sample teachers, students, courses, and assignments.
5.4.1	Implement authentication and role-based access control (F01)	Trương Hiển Long	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); implementing session auth and Teacher/Student guards.
5.4.2	Implement course management (F02)	Huỳnh Trung Ngạn	4	Analogy: 2-day duration; standard CRUD API endpoints for courses.
5.4.3	Implement enrollment and roster management (F03)	Lê Đoàn Nhật Huy	4	Analogy: 2-day duration; API endpoints for adding/removing students in course rosters.
5.4.4	Implement course material upload and retrieval (F04)	Trương Hiển Long	8	Three-point estimate: O=6h, M=8h, P=10h -> E=8h (4-day duration); integrating Cloudflare R2 presigned URLs/buckets and metadata.
5.4.5	Implement assignment creation and submission (F05)	Huỳnh Trung Ngạn	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); handling deadline validation and student file/link submissions.
5.4.6	[Optional] Implement secondary features (F06-F09)	Unassigned (Optional)	0	Expert judgement: 0 baseline hours (Should/Could priority); implemented only if schedule buffer permits.
5.5.1	Verify the testing environment and test data	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; validating test database state before test execution.
5.5.2	Execute Unit Testing	Trương Hiển Long	2	Three-point estimate: O=1h, M=2h, P=3h -> E=2h (1-day duration); running Vitest suite and generating V8 coverage reports.
5.5.3	Document the test results and defects	Lê Đoàn Nhật Huy	2	Expert judgement: 1-day duration; logging pass/fail outcomes and tracking defects on GitHub Issues.
5.5.4	Set up automated system testing (at least 2 use cases)	Lê Đoàn Nhật Huy	4	Three-point estimate: O=3h, M=4h, P=5h -> E=4h (2-day duration); writing Playwright E2E tests for 2 main use cases.
5.6.1	Deploy the application to the deployment environment	Trần Hoàng Minh Khang	4	Three-point estimate: O=2h, M=4h, P=6h -> E=4h (2-day duration); deploying Nuxt 4 bundle to Cloudflare Pages/Workers and verifying R2/DB.
5.7.1	Verify that the application is working as expected	Trần Hoàng Minh Khang	2	Expert judgement: final production smoke testing on live Cloudflare URL.
5.7.2	Document the project retrospective	Huỳnh Trung Ngạn	2	Expert judgement: evaluating plan-vs-actual effort, schedule, and lessons learned.